

Seunggeun Kim

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Summary

I am a third-year Ph.D. student in Electrical and Computer Engineering at The University of Texas at Austin. My research focuses on sequence modeling with diffusion models and their real-world applications across diverse domains, including protein design, natural language processing, and code generation. I am also interested in uncovering the latent reasoning capabilities of language models to enable more efficient and human-like reasoning.

Education

- PhD** **University of Texas at Austin**, Electrical and Computer Engineering Aug 2023 – Present
- Advisor: Adam Klivans
- BS** **Seoul National University**, Electrical and Computer Engineering Mar 2017 – Feb 2023
- GPA: 3.9/4.0 (summa Cum Laude)
 - 2 years absence to fulfill military duty

Publications

[From Interface to Inference: Eliciting Any-Order Inference from Any-Order Models](#), Submitted to NeurIPS 2026

Seunggeun Kim*, Jaeyeon Kim*, Taekyun Lee*, Yuyuan Chen*, Yilun Du, Sham Kakade, Sitan Chen

[Fine-Tuning Masked Diffusion for Provable Self-Correction](#), **ICML 2026**

Seunggeun Kim*, Jaeyeon Kim*, Taekyun Lee*, David Z. Pan, Hyeji Kim, Sham Kakade, Sitan Chen

[PPAAS: PVT and Pareto Aware Analog Sizing via Goal-conditioned Reinforcement Learning](#), **ICCAD 2025**

Seunggeun Kim, Ziyi Wang, Sungyoung Lee, Youngmin Oh, Hanqing Zhu, Doyun Kim, David Z. Pan

[Addressing Continuity and Expressivity Limitations in Differentiable Physical Optimization: A Case Study in Gate Sizing](#), **ISEDA 2025**

Yufan Du, Zizheng Guo, Yang Hsu, Zhili Xiong, **Seunggeun Kim**, David Z Pan, Runsheng Wang, Yibo Lin

[DICE: Device-level Integrated Circuits Encoder with Graph Contrastive Pretraining](#), **Under Review**

Sungyoung Lee, Ziyi Wang, **Seunggeun Kim**, Taekyun Lee, Yao Lai, David Z. Pan

[M3: Mamba-assisted Multi-Circuit Optimization via MBRL with Effective Scheduling](#), **ICCAD 2025**

Youngmin Oh, Jinje Park, **Seunggeun Kim**, Taejin Paik, David Pan, Bosun Hwang

[LLM-enhanced Bayesian optimization for efficient analog layout constraint generation](#), **arXiv preprint**

Guojin Chen, Keren Zhu, **Seunggeun Kim**, Hanqing Zhu, Yao Lai, Bei Yu, David Z Pan

Experience

Samsung AI Center, Research Intern

- Mentor: Doyun Kim
- Reinforcement Learning for PVT-aware analog sizing

Suwon, Korea
May 2024 – Jul 2024

Seoul National University, Research Intern

- Advisor: Jaeha Kim
- Project lead of DSP tape-out with Samsung 14nm FinFET technology

Seoul, Korea
Jan 2021 – Jul 2023

Honors and Awards

DAC Young Fellowship	2024
UT Austin Engineering Fellowship	2023

Technologies

Programming Languages: Python, Verilog, C, C++, Java

Technologies: Pytorch, Tensorflow, Linux, GIT

Circuit Tools: Virtuoso, Spectre, Design Compiler, Vivado, XMODEL, NGSpice